

Chapter

AR-Edge: Autonomous and Resilient Edge Computing Architecture for Smart Cities

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Abstract

With the rapid advancements in artificial intelligence (AI), the Internet of Things (IoT), and network communication technologies, recent years have witnessed a boom in smart cities that has dramatically changed human life and society. While many smart city applications rely on cloud servers, enabling comprehensive information fusion among users, smart devices, and service providers to provide diverse, intelligent applications, IoT networks' high dynamicity and heterogeneity also bring performance, security, and interoperability challenges to centralized service frameworks. This chapter introduces a novel Autonomous and Resilient Edge (AR-Edge) computing architecture, which integrates AI, software-defined network (SDN), and Blockchain technologies to enable next-generation edge computing networks. Thanks to capabilities in terms of logically centralized control, global network status, and programmable traffic rules, SDN allows for efficient edge resource coordination and optimization with the help of artificial intelligence methods, like large language models (LLM). In addition, a federated microchain fabric is utilized to ensure the security and resilience of edge networks in a decentralized manner. The AR-Edge aims to provide autonomous, secure, resilient edge networks for dynamic and complex IoT ecosystems. Finally, a preliminary proof-of-concept prototype of an intelligent transportation system (ITS) demonstrates the feasibility of applying AR-Edge in real-world scenarios.

Keywords: edge computing, internet of things (IoT), artificial intelligence (AI), security, Blockchain, software defined networks (SDN), smart cities, internet of vehicles (IoV), intelligent transportation systems (ITS)

1. Introduction

With the proliferation of the Internet of Things (IoTs) atop the fifth-generation and beyond (5G) communication technology, tens of billions of physical devices with network connectivity allow for a big band of data. Thanks to fast advancements in artificial intelligence (AI) and big data technology, recent years have witnessed a boom in ubiquitous and sustainable applications deployed on powerful cloud servers to link heterogeneous IoT devices through the Internet. As a result, Smart Cities have become realistic, dramatically changing human life and society by constructing

intelligent, sustainable, and safe living environments [1, 2]. However, with the exponential increase of physical devices and continuous development of diverse smart applications, a conventional system architecture that is solo based on the cloud computing paradigm nevertheless encounters many problems in efficiently handling the massive IoT data, satisfying Quality of service (QoS), and providing security and interoperability guarantees demanded practical scenarios in high dynamic and distributed network environments.

By migrating partial computing, storage, and networking capabilities from centralized cloud servers to the network edge near the end users, edge computing has emerged as a promising paradigm to meet challenges on cloud-centric IoT applications, like reducing end-to-end latency and improving security and privacy [3]. With the breakthroughs in AI, especially for machine learning (ML) techniques, integration of ML with edge computing has become an inevitable trend in the data-driven intelligent applications brought by IoT. Unlike transitional AI applications relying entirely on cloud computing, edge intelligence [4] allows most of the distributed edge computing resources to achieve intelligent capabilities to support diverse user-defined services and applications in Smart Cities. Because edge computing acts as an intermediary service layer between physical devices and intelligent services on cloud servers, a hierarchical cloud-edge computing architecture is widely adopted by large-scale and complex IoT-based applications [5].

The smart applications atop edge intelligence and IoT networks have numerous benefits, such as ultra-low latency, flexibility, robustness, and privacy preservation [6]. Nonetheless, due to the inherently dynamic and distributed nature of traditional IoT-Edge networks, centralized service frameworks still face significant challenges in performance, scalability, security, and privacy. Thanks to crucial characteristics like decentralization, immutability, and tractability, Blockchain has demonstrated great potential to revolutionize various aspects of the economy and society. Integrating blockchain and edge computing into one system is promising to provide decentralized management and trustworthy services for dynamic and distributed IoT networks [7]. As an intelligence-enhancing enabler in 5G networked systems, the Software Defined Network (SDN) promises to apply ML techniques to the heterogeneous network infrastructure. Thanks to capabilities in terms of logically centralized control, global network status, and programmable traffic rules, SDN empowered with ML methods allows for efficiently organizing, managing, maintaining, and optimizing resources (e.g., computing, storage, and networking) within multi-dimensional and self-autonomous networked systems [8].

1.1 Main contributions

This chapter introduces a novel Autonomous and Resilient Edge computing network architecture called AR-Edge to enable next-generation edge computing networks (NextG Edge). AR-Edge is a secure-by-design system infrastructure that integrates Blockchain, SDN, edge computing, and AI/ML technology to meet the challenges of current IoT-based ecosystems. **Figure 1** demonstrates a conceptual architecture of AR-Edge for Smart Cities, which acts as a backbone framework to link heterogeneous IoT devices and complex smart application domains. The multiple pervasively deployed IoT devices (e.g., sensors, cameras, and smartphones), network devices, and databases construct a fundamental physical infrastructure that offers data and resources (computing, communication, and storage) for the essential functionality of applications.

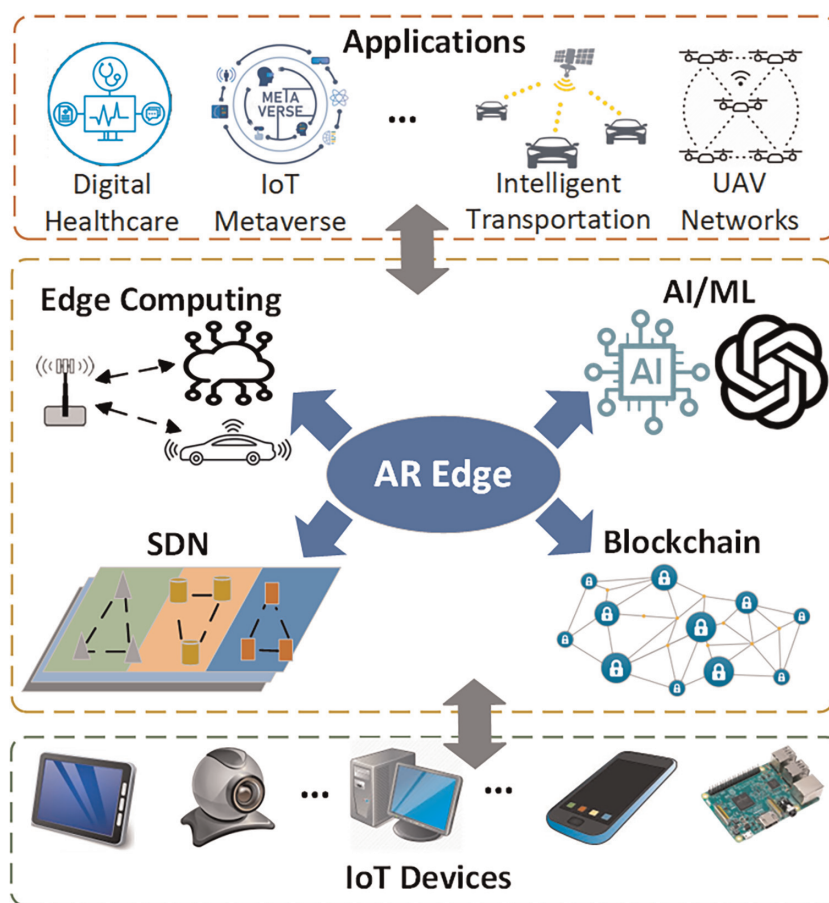


Figure 1.
 Conceptual Architecture of AR-Edge for Smart Cities.

In AR-edge, SDN controllers work as actuators that focus on performance improvements by handling heterogeneous IoT networks and dynamic resource allocations. The centralized SDN controllers can have a global view of the network by monitoring real-time network state and operational configuration. Thus, the huge volume of collected data can facilitate the applications of AI/ML techniques to enhance the whole system. As a secure network fabric for AR-Edge, Blockchain introduces decentralization to mitigate single point failure caused by centralized management and service frameworks; therefore, it can ensure system resilience and availability under distributed network environments. In addition, decentralized security mechanisms atop Blockchain can protect IoT devices, edge computing platforms, SDN controllers, and even AI/ML models against cyber threats, such as Sybil attacks, unauthorized access to devices, modifying training data, privacy violations, and distributed denial of service (DDoS).

In AR-edge, edge computing technology is the cornerstone that benefits other core components by providing capabilities like computing, networking, and storage. By offloading ML training and reference tasks from central cloud servers to the distributed edge servers that are close to IoT devices, the response time of decision-making, data transmission delay, and network bandwidth cost can be dramatically reduced. In addition, edge computing also provides virtualization and softwarization platforms

that can manage physical devices and SDN controllers in efficient and standard manners. Moreover, edge computing platforms also offer resources for Blockchain networks, like computation required by miners or validators and storage for saving distributed ledgers. By leveraging powerful AI/ML methods, like large language models (LLM), AR-Edge relies on a global intelligent agent as the system brain to achieve a self-autonomous (self-adaptive, self-healing, and self-organizing) network infrastructure. Finally, AR-Edge aims to support various and multiple instances of IoT applications in Smart Cities (such as digital healthcare [9], IoT-enabled Metaverse [10], connected intelligent vehicles, and urban air mobility systems [11]).

The rest of this chapter is organized as follows. Section 2 describes fundamental concepts of Edge computing and AI integration. Section 3 introduces the general architecture and workflow of SDN and explains how SDN can be applied in AR-Edge to guarantee performance requirements. Section 4 explores emerging Blockchain technologies for AR-edge regarding security and privacy preservation. Section 5 presents a practical internet of vehicle example to illustrate how the proposed AR-Edge can be applied in real-world applications, like intelligent transportation systems (ITS) in smart cities. Finally, Section 6 provides a summary and discusses the open challenges and future directions.

2. AI-enabled edge intelligence for IoT

2.1 Overview of edge computing

The Internet of Things (IoT) concept was first introduced to the community in 1999 for supply chain management [12]. IoT enables a computer system to “feel” contextual information and respond with actions without human intervention. Therefore, it has greatly increasingly permeated human lives and become an essential enabling technology in Smart Cities. Due to powerful capabilities in computation and storage, the cloud computing paradigm is widely used to construct scalable service infrastructures that collect data from distributed IoT devices and provide services and applications for users in a centralized manner. In recent years, this world has witnessed the proliferation of IoT and cloud computing technology that connect users, applications, and devices through the Internet and continuously promotes a safe community and sustainable society [13]. With the continuous development of IoT devices, the massive growth of data, and various QoS requirements of applications, cloud computing-based infrastructures show many shortcomings, such as not providing a real-time response, privacy leakage, and high energy consumption [14].

As a new enabling technology that allows computation to be performed at the edge of the network, edge computing refers to computation and network resources along the path between data sources and cloud servers; downstream data represents cloud servers, and upstream data represent IoT devices [15]. Unlike the cloud computing model that collects data produced by IoT and then consumes them on the cloud servers, the edge computing model makes data aggregated and used (consumed by services) at the edge of the network. By migrating powerful computing and network capabilities and rich storage resources from cloud servers to the edge of the network, edge computing has various characteristics to serve intelligent services and critical applications based on distributed IoT networks. For example, the closer data source leads to low latency computing, the reduced network bandwidth usage achieves efficient energy consumption, transferred computing power to improve QoS and user experience in time-sensitive applications [16].

2.2 The convergence of AI and edge computing

Facilitated by the advancements of computing capabilities (e.g., hardware and software) and big data processing techniques, AI, especially for deep learning (DL), has achieved unprecedented success in various application domains, such as computer vision, autonomous driving, and natural language processing. The cloud data center uses unlimited storage to aggregate and save the massive amount of heterogeneous data transmitted from IoT devices and sensors. A cloud server provides ubiquitous on-demand computing capability for these computing-intensive smart services and applications. Due to the high overload of data transmission on network bandwidth, inherent latency constraints of network communication, and the risks of leaking private and sensitive information during data analysis, the cloud-centric framework is not suitable for time-critical and privacy-sensitive applications [17].

Thanks to many advanced features, such as low latency, reduced bandwidth consumption, energy efficiency, and privacy protection, edge computing promises to solve the above issues by moving the computational capability closer to the information-generation source. **Figure 2** demonstrates the convergence of AI and Edge computing from the perspective of a hierarchical IoT-Edge-Cloud paradigm. From a system architecture aspect, edge intelligence (EI) acts as a service infrastructure layer to connect heterogeneous physical devices and high-level tasks running on cloud servers. As a middleware layer between the IoT stratum and the edge stratum, the communication and virtualization layer leverages abstraction and softwarization techniques to manage virtual resources by mapping physical devices to virtual resources according to their capabilities, such as computation, connectivity, and storage. Therefore, the edge intelligence framework uses virtual resources to manage physical devices developed on different hardware platforms and connected via diverse communication protocols efficiently. The cloud server works as a global “brain” to store system-level knowledge databases and manage global trained model aggregation and reference. Thanks to comprehensive knowledge and advanced ML algorithms on cloud servers, intelligent collaboration can support resource orchestration and service adjustment on the EI layer to handle dynamic and complex system condition changes.

The critical function blocks in EI can be classified into four groups: network control, security, resource management, and AI/ML model. By applying ML methods to global network data, edge intelligence empowers learning capability to SDN controllers that provide intelligent network control to satisfy QoS given the complex network environments. As an essential function in network control, traffic classification allows SDN controllers to identify various traffic flows and perform fine-grained network management. Supervised and semi-supervised learning methods are widely used for traffic classification, which can be divided into elephant flow-aware traffic classification, application-aware traffic classification, and QoS-aware traffic classification [8]. Routing optimization is another important function in network control by enabling SDN controllers to modify flow tables in switches to achieve the optimal routing of traffic flows. By using a supervised learning algorithm called long short-term memory (LSTM) to estimate future network traffic, NeuRoute can calculate the optimal heuristic-like routing solutions in real-time [18]. Thanks to capabilities to solve decision-making problems, reinforcement learning (RL) algorithms are used to develop a distributed intelligent routing protocol, which allows SDN controllers to select optimal data transmission paths given the network status [19].

Both cryptographic mechanisms and Blockchains are widely used in edge intelligence to provide security and privacy-preserving guarantees. Encryption methods can

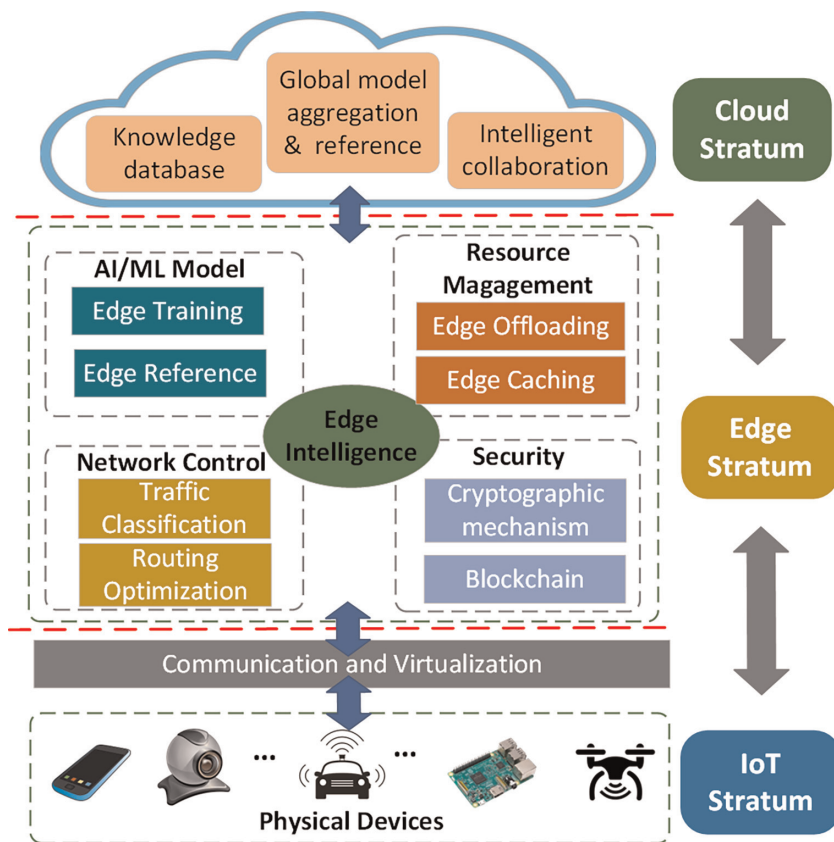


Figure 2.
The hierarchical framework of edge intelligence for IoT systems.

be used to ensure confidentiality during data transmission. Homomorphic encryption methods can especially protect private information in training and reference. In addition, asymmetric encryption and digital signature can provide identification authentication and access authorization for physical devices and virtual resources management. As a fundamental technology to ensure trust and decentralized network infrastructure, Blockchain can be integrated into EI to achieve dynamic resource orchestration and service re-adjustment for multi-domain IoT ecosystems [20].

As an extension of the cloud computing paradigm, edge computing offers infrastructure containing computing, storage, and network resources for edge training and edge reference. Edge training adopts a distributed learning architecture, like federated learning (FL), to learn the optimal values for all the weights and biases of DL models and identify hidden system patterns by training data sets stored at edge servers or IoT devices. For example, EI can provide a hierarchical FL framework to aggregate local training models into a global training model on the cloud server without directly exposing sensitive information of training data on devices [21]. Regarding high-quality EI service development, globally trained models and algorithms on cloud servers can be distributed at edge servers and devices to achieve high accuracy and low computation cost on testing instances.

Moreover, EI also provides resource management to support critical elements in intelligent applications and tasks, like training data collection, ML/DL model deployment, and computation provision. As EI leverages the edge computing paradigm to link distributed applications proximity to end-users and geographically scattered physical devices that record environmental information, edge caching techniques can collect the data generated from physical devices, such as networked cameras and sensors. The data are stored at reasonable places (edge servers or devices) and used for processing and analysis by intelligent algorithms to provide services for end-users [6]. For example, the raw video captured by cameras could be cached on edge servers for smart surveillance systems [22]. As an essential component of resource management in EI, edge offloading can leverage well-organized computing resources across the edge network to offer computing services for EI, like edge training and reference and network control.

3. SDN-based network control

3.1 SDN overview

The growing integration of edge devices has significantly increased the number of network management devices deployed to accommodate the traffic demand. Traditional network architecture relied on protocols such as Simple Network Management Protocol (SNMP), widely used to monitor network statistics and manually deploy network changes. Along with SNMP, NETCONF is another protocol more commonly used to automate the configuration of network devices. Although the protocols were available for remote configuration of networked devices and actively managing the traffic, the modern data throughput has significantly increased the bandwidth consumption closer to edge nodes and cloud data centers.

Integration of network devices from multiple manufacturers has increased the complexity of vendor-specific configurations, high cost, custom programming languages, and protocols specific to certain devices. For a constantly growing network connectivity, incompatibility among the networked devices could disable the network architecture functionality. To improve the interoperability among different vendors and create a network architecture capable of adapting to dynamic changes demanded by modern internet traffic, an efficient networking solution named Software Defined Networking (SDN) was devised.

The SDN has introduced a paradigm shift in the computer network architecture by decoupling the control plane and the data plane of the network devices. The data plane devices, such as the switches and the routers, are thereby responsible for network packet forwarding from interface to interface based on the instructions provided by the controller. The control plane maintains the bird's eye view of the network domain that the controller manages and provides user-instructed packet forwarding instructions to the data plane devices. The SDN consists of software-defined controllers that leverage Application Programming Interfaces (APIs) for managing the data plane hardware and leverage standardized protocols such as OpenFlow for their management [23]. The networked devices manufacturing industry has standardized the open communication channels among their devices using protocols such as OpenFlow, which enables remote changes to the device configurations using an SDN Controller. The integration of SDN has improved the interoperability among multiple vendor-networked devices and enabled dynamic changes to the link connectivity among the devices [24].

3.2 SDN architecture

As a result, the capability of separating the control plane and the data/forwarding plane has enabled SDN with dynamic and efficient programming of the network connections through a centralized controller hub [25]. To manage the devices in the forwarding plane, the SDN leverages interfaces through network connectivity. There are two dedicated application interfaces named Northbound and Southbound Interfaces. The Northbound Interface connects the SDN controller to the management plane where the network applications dedicated to the incoming network packets are running. Multiple network applications, such as firewall, tunneling, packet forwarding, load balancing, etc., are leveraged using the northbound interface. The Southbound interface connects the controller to the forwarding switches in the data plane to control the forwarding device, such as physical, wireless, optical, and virtual switches. The forwarding devices are controlled through supported network switches like open switches. However, it is not limited to that any forwarding device with supporting management protocols like OpenFlow, OVSDb, NetCONF, and SNMP can be controlled.

The SDN controller serves as the core of the control plane. Most SDN controllers cater services such as Topology, Inventory, Statistics, and Host tracking services. The topology service enables the discovery of the forwarding devices and their connectivity to other devices by using the Link Layer Discovery Protocol (LLDP) packets sent by the switch and observing the packet trajectory. The inventory service allows the SDN controller to track and record all SDN-enabled devices and their supported capabilities, like Openflow support. The statistics service reads counter information of the forwarding devices by using the flow table entries. Finally, the host tracking service discovers where the IP or MAC addresses are located on the network's topology.

Figure 3 represents the SDN architecture with individual components. With the SDN Controller established with access to the network applications on the Northbound interface and the forwarding devices on the Southbound interface, the controller can now manage the network traffic based on user specifications. For example, when a new host is connected to the switch in the data plane and tries to communicate with other devices in the network, the switch first consults the controller on instructions to handle the incoming packet from the new host. Based on the destination IP/

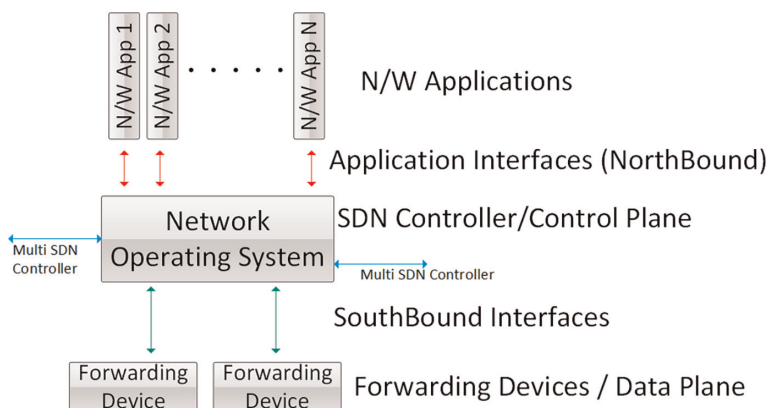


Figure 3.
SDN Architecture with North and Southbound Interfaces.

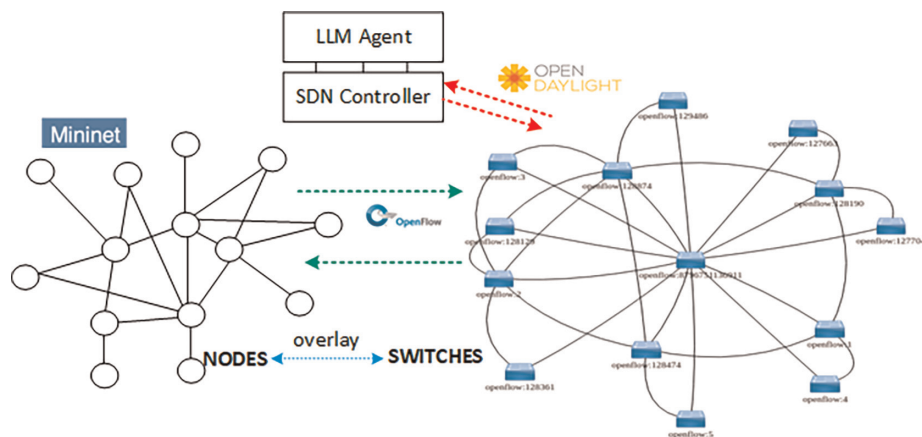


Figure 4.
 SDN testbed initialized with Mininet and OpenDaylight SDN Controller.

MAC, the SDN Controller updates the switch with flow table entries, including instructions on handling such packets in the future with a timeout. With each switch flow table entries updated, the new host can communicate seamlessly with other nodes, and the controller maintains the inventory accordingly.

As the network scale gets larger, the resources in the SDN controller get limited. The centralized approach for SDN to manage the whole network could limit its optimal functionality for large-scale networks. However, multiple SDN controllers deployed to manage sectional networks allow for optimal functionality. The controllers update their inventory based on their East/Westbound interfaces, and the user can increase the scale of the network with unique or duplicate network applications catered by each controller. **Figure 4** represents a simulated network in a Mininet environment [26], and the OpenDaylight (ODL) SDN controller discovers the resulting switches [27]. Each node represents an open switch and can update its flow table entries based on the instructions received by the ODL Controller.

Leveraging network applications like routing optimization [18] for high-traffic network scenarios along with load balancing the SDN controller utilization with load balancing techniques, the network functionality is resilient to dynamic demands. Paired with LLM agents for supervised network traffic prediction and optimal route calculation, the autonomous SDN controller allows for an optimal edge network architecture.

4. Blockchain enhanced security

4.1 Blockchain overview

As the underlying technology of Bitcoin [28], *Blockchain* has demonstrated excellent capabilities to revolutionize traditional information systems based on the centralized system architecture. Essentially, Blockchain is a distributed ledger technology (DLT) atop decentralized Peer-to-Peer networks such that transactions and blocks are stored as a verifiable, append-only chained data storage. Blockchain leverages basic cryptographic primitives, such as cryptographic hash functions, asymmetric/

symmetric encryption, digital signature, and access control, to offer security guarantees. For example, cryptographic hash functions are widely used in consensus algorithms (e.g., Proof-of-Work and Proof-of-Stake) and committee election [29]. From a network architecture, Blockchain is an overlay network system that relies on Peer-to-Peer (P2P) network technology, like gossip communication [30] and distributed hash table (DHT) protocols [31, 32], to propagate control messages, transactions, and blocks without a centralized coordinator. The P2P network offers reliability and scalability for Blockchain data transmission in a large-scale distributed network environment.

As an essential function of a Blockchain system, consensus protocols allow all participants (miners or validators) to agree on the distributed ledger that maintains data integrity, consistency, and order of data across the distributed network without any third-party authority. The consensus in a distributed system aims to solve the Byzantine General Problem [33], which requires a single value agreement among different system parts given the failure of communication or conflicting information. Regarding various consensus protocols, Blockchain can be classified as permissionless blockchain (e.g., PoW and its variant) and permissioned blockchain (e.g., Practical Byzantine Fault Tolerant (PBFT) and its variant) [34]. Due to good scalability and global security in an open-access network environment, PoW has been used by public blockchain networks like Bitcoin and Ethereum. However, PoW requires high computation resources in mining blocks, such that it incurs unsustainable electrical energy consumption. PBFT [35] demonstrates better performances than PoW, such as low block confirmation latency, high transaction throughput, and less computation and energy consumption. However, it needs identity authentication and allows for limited network scalability regarding the number of validators during the consensus process.

By leveraging cryptographic primitives and secure computing mechanisms of Blockchain, *Smart Contract* can be used to develop self-enforcing and self-executing programs, which actuate the term of rules of a particular agreement or contract [36]. Smart contracts bring programmability to Blockchain by integrating business logic and user interfaces, such as offering complex operations and flexible services rather than solo cryptocurrency and cash-by-cash payment. Through publishing a set of application binary interfaces (ABIs), smart contracts act as autonomous trust agents between parties to fulfill predefined contract agreements under specific conditions [37]. Therefore, smart contracts can be used to implement decentralized applications (DApp) for services and applications under distributed and trust-less network environments.

4.2 Blockchain-based security solutions to edge computing

Thanks to the distributed nature of the edge network and the security guarantees of Blockchain, the integration of Blockchain and edge computing promotes decentralized, secure, and reliable EI services and applications.

Blockchain can be integrated into the EI system to ensure data transmission between physical devices, SDN controllers, and edge servers. For example, a blockchain-based distributed cloud architecture enables SDN-based fog nodes to interact with each other and through a blockchain network, and it can provide low-cost, secure, and on-demand access to the computing resources in a distributed edge computing network [38]. DistBlockNet [39] integrated Blockchain into a SDN-based edge computing network to update the flow rules table via SDN controllers, such switch devices can securely verify and validate flow rules table as downloading them.

Blockchain can be applied to the EI system to improve the capacity and security of data storage at edge computing networks. Due to the diverse formats and sizes of raw data generated by EI, Blockchain cannot directly store these data on the distributed ledger. Thus, hybrid on-chain and off-chain storage have been adopted by many Blockchain-based decentralized data storages for IoT applications [40]. All raw data are saved into off-chain storage, which is implemented by InterPlanetary File System (IPFS) [41] or Swarm [42]. While the metadata and reference of raw data are stored in transactions committed on Blockchain for verification during data accessing and sharing. Thus, Blockchain and decentralized storage can provide reliable, traceable, and tamper-proof data services without sacrificing security and privacy preservation.

5. Case study: towards security and resilience of intelligent transportation system

With the development of communication, network, and mobile computing technology, vehicles equipped with smart devices, such as wireless sensors, onboard computers, GPS antennas, cameras, radar, and so on, can collect and process large amounts of context-aware data while enabling information exchange between vehicles [43]. Thanks to advancements in vehicular communication and self-driving technology, Internet of Vehicles (IoV) becomes realistic through seamless interconnection among smart vehicles, roadside infrastructure, pedestrians, transportation service providers, and intelligent traffic management systems. By enabling a comprehensive information exchange platform between highly connected smart vehicles and heterogeneous vehicular services, an intelligent transportation system (ITS) leverages AI and big data techniques to provide diverse intelligent and safe vehicular applications, such as enhanced pedestrian and driving safety, efficient traffic planning, smart parking, and entertainment services [44].

The ITS aims for a seamless, connected, and ubiquitous service platform that supports large volume data collecting, transacting, and sharing among participants such as vehicles, pedestrians, and service providers. Meanwhile, existing ITS systems that rely on cloud-based storage and management incur new concerns on performance, scalability, interoperability, security, and privacy. First, ever-increasing interconnected vehicles, along with a massive amount of vehicular data, introduce scalability issues in the centralized ITS services framework. In addition, future ITS also considers interoperability as sharing data and resources with a wide range of service providers, such as original equipment manufacturers (OEM), insurance companies, and transportation departments. Because ITS uses advanced AI/ML algorithms and models to provide diverse smart applications, edge computing nodes allow for caching data from vehicles and offloading tasks from cloud servers to improve QoS. Moreover, the centralized frameworks adopted by ITS are prone to performance bottlenecks and single points of failure under highly dynamic and distributed network environments. Therefore, it is necessary to rethink the system architecture for next-generation IoV networks and intelligent service platforms.

5.1 Design rationale and system architecture

To address the aforementioned issues in current IoV ecosystems, a novel system architecture is introduced by integrating the AR-Edge framework with multi-domain IoV networks to ensure the security and resilience of ITS. **Figure 5** demonstrates a

system architecture of interconnected vehicular service networks based on a crossroad scenario. The whole IoV network consists of four independent and fragmented vehicle networks. Each vehicle network adopts an AR-Edge framework to perform domain-specific network control, resource orchestration, intelligent services deployment, and security and privacy enforcement. As a permissioned network, each vehicle network relies on system administration services deployed on edge servers to manage all registered entities within a vehicle network. As **Figure 5** shows, these entities could be vehicles, cameras, traffic lights, roadside unit (RSU), smart devices used by pedestrians, communication infrastructure, edge servers, charging stations, etc. Each registered entity within a vehicle is assigned a unique identifier (UID) by a trust system administrator.

ITS ecosystems use IoV networks and computing technology to enable information and data sharing among participants and achieve a self-organized network. The information interaction between vehicles and other entities refers to vehicle-to-everything (V2X) models, which include vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-roadside unit (V2R), vehicle-to-grid (V2G), and vehicle-to-pedestrian (V2P) [43]. Each vehicle uses onboard control units and deployed sensors (e.g., inertial measurement unit (IMU), radar and camera, etc.) to collect and process car operating states and environmental information. After joining an IoV network, the vehicle can use an embedded telematics control unit (TCU) to offer wireless communication to and from vehicles and other entities. V2V allows vehicles to broadcast useful information to each other, such as emergency braking, collision detection,

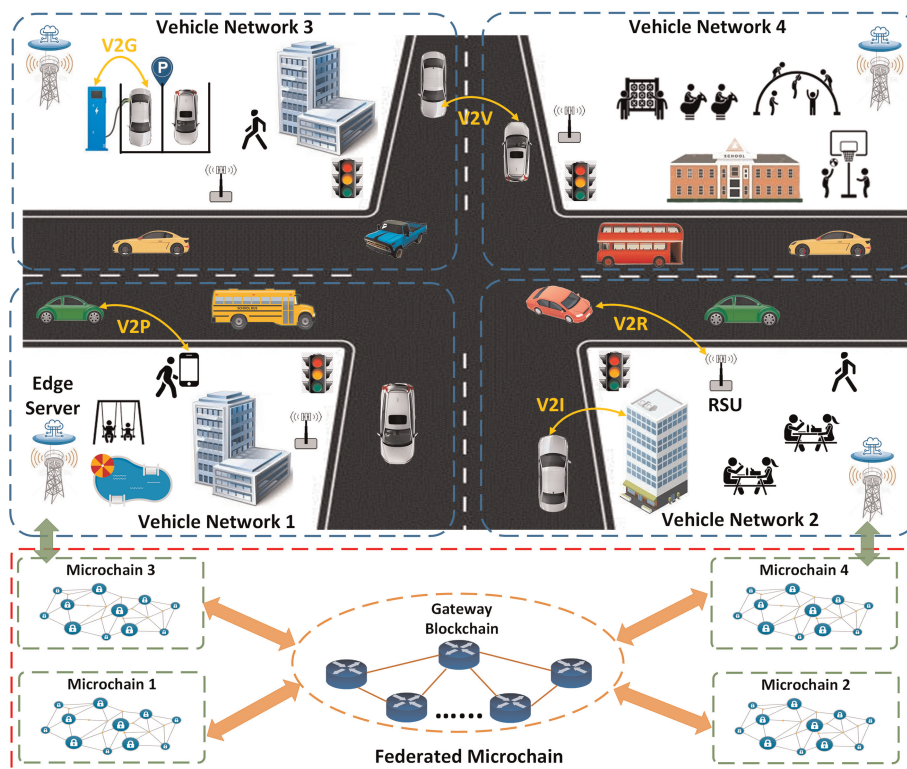


Figure 5.
The system overview of AR-Edge-enabled IoV networks.

and road traffic conditions, thereby improving user driving safety and travel efficiency [43].

In an IoV network, RSUs are placed along the roadside to support V2R and V2I for information exchange between vehicles and other network infrastructures. The RSUs are edge computing nodes that provide storage to save vehicle data and traffic information within a vehicle network. In addition, deploying network control functions on RSUs can improve network resource allocation and interoperability between vehicle communication and heterogeneous service and application networks. Furthermore, RSUs can perform local decision-making based on real-time traffic conditions to offer optimal vehicle travel routines within the covered service range. By collecting vehicle data and traffic information, the edge server of a vehicle network acts as the “smart brain” that maintains the global traffic status and performs global traffic strategy. The cloud server processes computation-intensive LLM training tasks on multidimensional transportation system data (vehicle data, road traffic information, network states, and services deployment.) Given individual vehicle network status and QoS requirements, a fine-tuned model can be deployed on the edge server that performs offloading intelligent tasks to optimize transportation service provision and ensure public safety for drivers and pedestrians.

As a decentralized and trust-free network infrastructure for ITS, a federated microchains framework [20] is integrated into IoV ecosystems to ensure reliable and secure data sharing and resource allocation across multiple vehicle networks. The existing Blockchain-based IoV solutions rely on monolithic Blockchains such that they cannot handle the Blockchain trilemma [45]. However, the hierarchy of the federated microchains promises to make trade-offs in decentralization, efficiency, scalability, and security by applying blockchain to dynamic and heterogeneous IoV networks. As the bottom of **Figure 5** shows, each vehicle network relies on a microchain to record data and transactions within the network. At the same time, a global gateway Blockchain links multiple fragmented microchains. In each microchain, a periodically randomly selected committee executes a lightweight consensus protocol, such as PBFT or Delegated Proof-of-Stake (DPoS), to verify and store data on a private distributed ledger. In addition, a microchain leverages a hybrid storage scheme by combining an on-chain ledger and an off-chain distributed database to guarantee the security and privacy of sensitive data within a vehicle network.

From a global security aspect, delegating nodes selected from vehicle networks construct a global gateway blockchain to guarantee scalability and interoperability for multi-domain IoV networks. The gateway Blockchain relies on a PoW consensus protocol to maintain a publicly distributed ledger. For multi-domain operations, raw data are saved on permissioned microchains, while references or metadata of original data are saved into checkpoint blocks that are finalized on the gateway Blockchain. Therefore, gateway nodes can utilize inter-microchain protocols that rely on global checkpoint blocks to ensure auditability, immutability, and provenance for cross-microchain transactions.

5.2 Prototype implementation and evaluation

To verify the feasibility of integrating AR-Edge with IoV networks, a proof-of-concept prototype is implemented with Python and tested on a virtual network environment by using Mininet [26]. The network topology of the testbed consists of two virtual IoV networks and one virtual gateway blockchain network, all deployed on an HPC. All virtual networks are connected through a remote controller deployed on a

	HPC-Workstation	Desktop
CPU	2.2GHz, Intel(R) Gold 5520R (96 cores)	3.4GHz, Core (TM) i7-2600 K (8 cores)
Memory	512GB DDR3	16GB DDR3
Storage	4 TB HDD	500GB HDD

Table 1.
Configuration of experimental devices.

desktop. **Table 1** describes devices used for the experimental study. This case study evaluates how a microchain federation can improve the security and interoperability of multidomain IoV ecosystems. Therefore, a private Ethereum Blockchain [46] is set up on one virtual gateway blockchain network, while the private Tendermint blockchain [47] is configured on two other virtual IoV networks. Multiple virtual hosts are created for each virtual network to simulate entities of IoV networks or blockchain gateway nodes, and each node is assigned one cup core.

The test cases are developed to evaluate the latency and throughput of processing transactions under test scenarios: query inter-ledger transactions and commit inter-ledger transactions given varying gateway nodes and system transaction throughput. The number of gateway nodes ranges from 4 to 20. The system transaction throughput Th_S is defined as “users send” transactions per second (tps) during query and commit data operations. The processing transaction throughput Th_P is denoted as the actual “system can process” transactions per second (tps) during inter-ledger operations. We conducted 50 Monte Carlo test runs for each case scenario and used the average of the results for evaluation.

First, we set four gateway nodes for each virtual network and evaluate latency and throughput by increasing Th_S from 20 to 1000 tps. **Table 2** shows the total processing latency of transactions as scaling up Th_S . Given a fixed number of gateway nodes, the end-to-end delays incurred by query operations are almost linear to Th_S . Because they are dominated by system capabilities, such as each host’s computing power, network link bandwidth, etc. The underlying Blockchain’s properties, like block confirmation time, have significant impacts on the latency of committing transactions on the distributed ledger. As a result, the end-to-end delays caused by commit operations are almost stable when $Th_S \leq 200$ tps. **Figure 6** plots trends of processing transaction throughput Th_P as scaling up Th_S . When $Th_S \leq 200$ tps, Th_P is almost linear to Th_S due to the stable latency of committing inter-ledger transactions. However, system capability becomes the performance bottleneck as $Th_S \geq 200$ tps such that Th_P becomes saturated. Th_P of query inter-ledger transactions demonstrates almost stable because the capacity of gateway nodes mainly influences them.

To evaluate how the number of gateway nodes influences performance, we fixed $Th_S = 1000$ for both query and commit scenarios while increasing the number of gateway nodes from four to 20. **Table 3** shows the total processing latency of

TPS	20	50	100	200	500	1000
Query inter-ledger transactions (second)	0.1	0.2	0.5	1.0	2.3	4.5
Commit inter-ledger transactions (second)	1.6	1.6	1.6	1.6	3.0	5.5

Table 2.
The latency as scaling up system transactions.

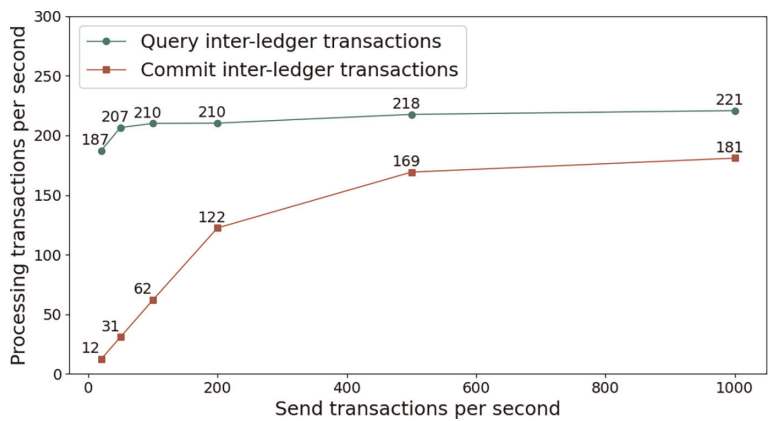


Figure 6.
Comparison of the throughput as scaling up transactions.

Number of nodes	4	6	8	10	12	16	20
Query inter-ledger transactions (second)	1.8	1.2	0.9	0.7	0.6	0.5	0.4
Commit inter-ledger transactions (second)	2.9	1.6	1.6	1.6	1.6	1.6	1.6

Table 3.
The latency as scaling up gateway nodes.

transactions as scaling up gateway nodes. AR-Edge can introduce resource allocation to dynamically configure gateway nodes and network control to coordinate transactions within the gateway blockchain network efficiently. Therefore, adding more gateway nodes to distributed service overload can dramatically reduce the total latency of processing a large volume of transactions in a short time period, especially for query operations. **Figure 7** demonstrates trends of processing transaction throughput Th_p as scaling up gateway nodes. Th_p of query inter-ledger transactions are almost linear to increase the number of gateway nodes. However, the performance

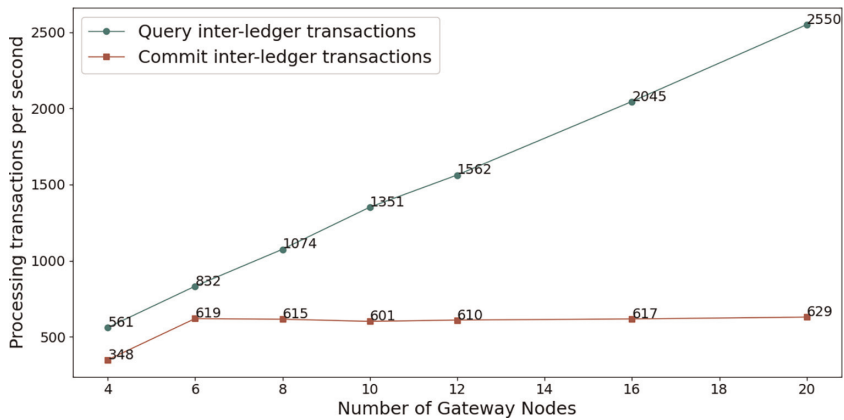


Figure 7.
Comparison of the throughput as scaling up gateway nodes.

of inter-ledger transactions is mainly dominated by the characteristics of gateway Blockchain using a PoW consensus protocol. Thus, solo increasing the number of gateway nodes cannot reduce processing latency or improve Th_P when the system is handling a large volume of transactions within a short time (e.g., $Th_S = 1000$). Applying a lightweight consensus protocol with short transaction committed time and high processing throughput into the gateway blockchain is a promising solution to improve system performance, especially for committing data on the distributed ledger.

6. Conclusions

This chapter introduces an AR-Edge that adopts AI/ML, edge computing, SDN, and Blockchain as enabling technologies to construct the next edge computing networks for diverse applications in smart cities. Given a comprehensive overview of AR-Edge architecture, we present a hierarchical framework of edge intelligence atop IoT ecosystems. We explain how core enabling technologies collaboratively promote an intelligent, secure, and resilient system architecture for dynamic and complex IoT scenarios. Finally, an ITS based on multiple IoV networks is demonstrated as a case study. The prototype analysis shows that AR-Edge is a promising services and applications framework in IoT ecosystems.

However, questions and open issues remain unanswered, such as designing AR-Edge in real-world scenarios and validating system performance and security features. One challenge is the gateway blockchain architecture design that ensures efficiency, security, and interoperability for cross-micro chain operations, especially for cases that integrate cryptocurrency networks (e.g., bitcoin and Ethereum) to support decentralized financial services. Another challenge is investigating AI/ML techniques to achieve intelligent SDN controllers that optimize resource allocations and enhance system security with proactive prediction and mitigation of cyber threats. Our ongoing efforts are developing an SDN-based edge intelligence framework that can be applied to practical applications, such as ITS atop a scalable IoV network and urban air mobility (UAM) system in smart cities.

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Abbreviations

AI	artificial intelligence
ABI	application binary interfaces
API	application programming interface
DApp	decentralized applications
DDoS	distributed denial-of-service
DHT	distributed Hash table
DL	deep learning
EI	edge intelligence
FL	federated

IMU	inertial measurement unit
IoT	Internet of Things
IoV	internet of vehicles
IPFS	interplanetary file system
ITS	intelligent transportation systems
LLDP	link layer discovery protocol
LLM	large language models
LSTM	long short-term memory
ML	machine learning
ODL	OpenDaylight
OEM	original equipment manufacturer
DPoS	delegated proof-of-stake
PoW	proof-of-work
P2P	peer-to-peer
QoS	quality-of-service
RSU	roadside unit
RL	reinforcement learning
SDN	software defined network
SNMP	simple network management protocol
TCU	telematics control unit
UAV	unmanned air vehicle
V2G	vehicle-to-grid
V2I	vehicle-to-infrastructure
V2P	vehicle-to-pedestrian
V2R	vehicle-to-roadside unit
V2V	vehicle-to-vehicle
V2X	vehicle-to-everything

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